Infrastructure as a Service (IaaS)

Infrastructure as a Service (IaaS) is a cloud computing service model that provides fundamental computing resources such as **virtual machines, storage, and networking** over the internet. Instead of purchasing physical hardware, users can rent these resources on demand from cloud providers.

In IaaS, the cloud provider manages the physical infrastructure (servers, data centers, networking hardware), while the user is responsible for managing the operating system, applications, and configurations.

IaaS acts as the **foundation layer of cloud computing**. It gives users maximum control over their environment without the need to maintain physical hardware. This makes it suitable for system administrators, developers, and organizations that require flexibility and customization.

How IaaS Works

Resource Provisioning

Users request infrastructure resources such as virtual machines or storage through a cloud platform. These resources are created instantly using virtualization technology.

Virtualization

Physical servers are divided into multiple virtual machines, each functioning as an independent system. This allows efficient use of hardware resources.

Deployment

Users install operating systems, software, and applications on the virtual machines based on their requirements.

Management

Users manage the system configuration, updates, security settings, and applications, while the cloud provider maintains the underlying hardware.

Key Components of IaaS

1. Compute (Virtual Machines)

Virtual machines act like physical computers where users can run operating systems and applications. These can be scaled up or down based on workload.

2. Storage

Provides different types of storage such as block storage (disks), object storage, and file storage for storing application data.

3. Networking

Includes virtual networks, IP addresses, load balancers, and firewalls that enable communication between resources.

Examples of IaaS Providers

- Amazon Web Services – EC2 (Elastic Compute Cloud)
- Microsoft Azure – Virtual Machines
- Google Cloud – Compute Engine

Use Cases of IaaS

Web Hosting

Organizations can host websites on virtual machines without buying servers.

Application Development and Testing

Developers can quickly create and test environments without setting up physical systems.

Data Storage and Backup

IaaS provides scalable storage for storing and backing up data.

Disaster Recovery

Organizations can replicate infrastructure in the cloud to recover quickly from failures.

Advantages of IaaS

- Provides full control over infrastructure
- Highly scalable and flexible
- No need to maintain physical hardware
- Cost-effective with pay-as-you-go pricing
- Quick deployment of resources

Disadvantages of IaaS

- Requires technical knowledge to manage systems
- Security responsibility is shared with the user
- Improper configuration can lead to vulnerabilities
- Cost management can become complex if not monitored

Platform as a Service (PaaS)

Platform as a Service (PaaS) is a cloud computing service model that provides a complete **development and deployment environment** in the cloud. It allows developers to build, test, deploy, and manage applications without worrying about the underlying infrastructure such as servers, storage, or operating systems.

In PaaS, the cloud provider manages the hardware and platform (including OS, runtime, and middleware), while the user focuses only on application development and data.

PaaS sits between IaaS and SaaS in the cloud service model hierarchy. It abstracts away infrastructure management and provides a ready-to-use platform with tools and frameworks required for application development.

This model is particularly useful for developers, as it reduces the complexity of setting up environments and accelerates the development lifecycle.

How PaaS Works

Platform Provisioning

The cloud provider sets up and manages the platform, including servers, operating systems, runtime environments, and development tools.

Application Development

Developers write application code using supported programming languages and frameworks provided by the platform.

Deployment

Applications are deployed directly onto the cloud platform without manual configuration of servers.

Management and Scaling

The platform automatically manages scaling, load balancing, updates, and maintenance, ensuring smooth application performance.

Key Components of PaaS

1. Development Tools

PaaS provides built-in tools, libraries, and frameworks that help developers write and test applications efficiently.

2. Runtime Environment

It offers execution environments where applications can run, such as Java, Python, .NET, or Node.js environments.

3. Middleware

Middleware connects applications with databases, messaging systems, and other services, simplifying integration.

4. Database Management

Many PaaS platforms include managed database services for storing and managing application data.

Examples of PaaS Providers

- Microsoft Azure – Azure App Service
- Google Cloud – Google App Engine
- Amazon Web Services – AWS Elastic Beanstalk

Use Cases of PaaS

Application Development

Developers can build applications quickly without setting up infrastructure.

Web Application Hosting

Web applications can be deployed and managed easily using platform services.

API Development and Integration

PaaS provides tools to create and manage APIs for connecting applications.

DevOps and Continuous Deployment

Supports automated testing, integration, and deployment pipelines.

Advantages of PaaS

- Simplifies application development and deployment
- Reduces the need for infrastructure management
- Faster development lifecycle
- Built-in scalability and high availability
- Supports multiple programming languages

Disadvantages of PaaS

- Less control over underlying infrastructure
- Platform dependency (vendor lock-in)

- Limited customization compared to IaaS
- May not support all programming environments

Software as a Service (SaaS)

Software as a Service (SaaS) is a cloud computing service model in which **software applications are delivered over the internet** on a subscription or usage basis. Users can access these applications through a web browser without installing or maintaining them on their local systems.

In SaaS, the cloud provider manages everything including infrastructure, platform, application software, updates, and security, while the user simply uses the application.

SaaS is the **top layer of cloud service models**, built on top of IaaS and PaaS. It provides fully functional software solutions that are ready to use without any setup or configuration.

This model is widely used because it simplifies software access and eliminates the need for installation, maintenance, and manual updates. Users only need an internet connection to access the service.

How SaaS Works

Application Hosting

The software application is hosted on cloud servers managed by the provider. These servers handle all processing and data storage.

User Access

Users access the application through a web browser or thin client. No installation is required on the user's device.

Data Processing

All processing and computations are performed on the cloud servers. The results are sent back to the user through the internet.

Automatic Updates

The provider manages updates, bug fixes, and security patches. Users always access the latest version of the software.

Key Characteristics of SaaS

1. Web-Based Access

Applications are accessed through web browsers, eliminating the need for installation.

2. Multi-Tenant Architecture

A single instance of the application serves multiple users (tenants), ensuring efficient resource utilization.

3. Subscription-Based Pricing

Users pay on a monthly or yearly basis instead of purchasing software licenses.

4. Automatic Maintenance

All maintenance, updates, and security are handled by the provider.

Examples of SaaS Applications

- Google Docs – Online document creation and editing
- Microsoft Teams – Communication and collaboration
- Gmail – Email service
- Zoom – Video meetings

Use Cases of SaaS

Business Applications

Used for enterprise tools such as CRM, ERP, and HR management systems.

Communication and Collaboration

Provides tools for messaging, video conferencing, and shared document editing.

Education and E-Learning

Used for online learning platforms, virtual classrooms, and course management.

Content Management

Helps in managing websites, blogs, and digital content.

Advantages of SaaS

- No installation or setup required
- Accessible from anywhere using the internet
- Automatic updates and maintenance
- Cost-effective subscription model
- Easy scalability for users

Disadvantages of SaaS

- Limited control over application features
- Dependency on internet connectivity
- Data security concerns in shared environments
- Vendor lock-in issues

Examples

1. IaaS – Amazon EC2 (Elastic Compute Cloud)

Amazon Web Services provides **EC2 (Elastic Compute Cloud)** as an IaaS service. It allows users to create and manage virtual machines (called instances) in the cloud.

What Service Provides

Amazon EC2 provides:

- Virtual servers (instances)
- Storage (EBS – Elastic Block Storage)
- Networking (security groups, IP addresses)

It essentially replaces physical servers with virtual infrastructure.

What User Manages

In EC2, the user is responsible for:

- Installing and managing the operating system (Linux/Windows)
- Installing applications and software
- Managing security configurations
- Handling updates and patches

How It Is Used

A company wants to host a web application:

1. Launch an EC2 instance
2. Install Linux OS
3. Install web server (Apache/Nginx)

4. Deploy application code
5. Configure database and networking

Why is it IaaS

Because:

- It provides only infrastructure
- User has full control over OS and applications
- Cloud provider manages only hardware

---

2. PaaS Example – Azure App Service

Overview

Microsoft Azure provides **Azure App Service**, a PaaS offering that allows developers to build and deploy web applications without managing infrastructure.

What Service Provides

Azure App Service provides:

- Pre-configured runtime environments (.NET, Java, Python, Node.js)
- Web server and hosting environment
- Built-in scaling and load balancing
- Continuous deployment support

What User Manages

In this case, the user focuses only on:

- Writing application code
- Managing application logic
- Handling data and configurations

How It Is Used

A developer wants to deploy a web app:

1. Write code in Python/Node.js

2. Upload code to Azure App Service
3. Configure environment settings
4. Application is automatically deployed

No need to install OS or manage servers.

Why It Is PaaS

Because:

- Platform (OS, runtime, server) is managed by provider
- User focuses only on development
- Infrastructure management is abstracted

---

3. SaaS – Google Docs

Google Docs is a SaaS application that allows users to create, edit, and share documents directly through a web browser.

What Service Provides

Google Docs provides:

- Complete document editing software
- Cloud storage integration
- Real-time collaboration features
- Automatic saving and version control

What User Manages

The user only:

- Creates and edits documents
- Shares files with others

No need to manage:

- Software installation
- Updates
- Infrastructure

How It Is Used (Example Scenario)

A student wants to create a report:

1. Open Google Docs in browser
2. Create a document
3. Type and edit content
4. Share with others for collaboration

Everything runs in the cloud.

Why It Is SaaS

Because:

- Fully functional software is provided
- No installation or setup required
- Entire system is managed by provider